MCR3M - M1C1

Agenda:

1. Minds-On: What is a Relationship?
2. Action: Functions & Notation
3. Formative Feedback: Portfolio Story Connection
4. Distinguishing Linear from Quadratic Functions
5. Consolidation: Exit Question

Learning Goal:

I will recognize functions from tables, graphs, and equations

Formative

For each and every class, as soon as you enter the Zoom room make sure you log into our class Formative page (bookmark it :)

(one-time only) Class code: **EW4J2J**

Open the page matching the module and lesson number

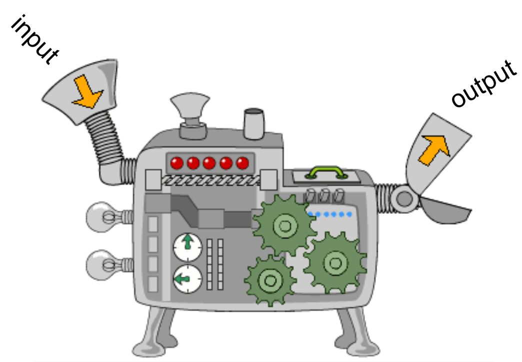
E.g. Today's class (the 1st class of Module 1) is **M1C1**

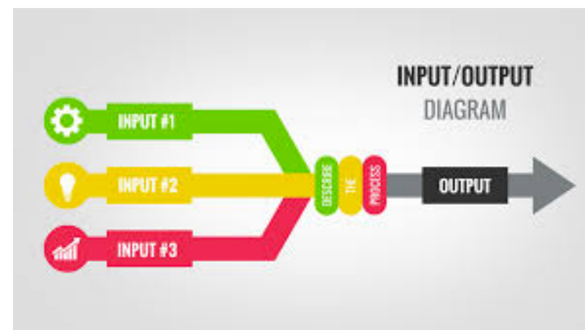
These days we have tools for calculating just about everything:

Your dog's human age

Asteroid Impact Effect

Behind the scenes, each of these tools utilizes some sort of mathematical computation that generates an estimate. In each problem, notice that we are considering many inputs to determine an output. This can be represented in many ways, including visually with pictures or diagrams:





Because this basic structure represents the same class of problem as many of the others that have been encountered over the years, it would make sense for us to create a standard system or a common language that we can use to model and classify the different ways that input and output can be related.

Fortunately this language already exists:

Functions

A **function** is just *well-behaving relationship*. To be more specific, ***when we pass unique input to a "function", we know that is only going to give us one output.***

Given a little thought, it makes sense that we'd want our tools top give us a specific unique output for every input; imagine how frustrating it would be if the dog's age calculator spit out different human age's for two people entering the same input information!

Mathematical Definition of a Function

A function is a relation that has exactly one output for every input

Ex. 1: Answer in Q1 GoFormative M1C1 Page - Do you think this is a function?

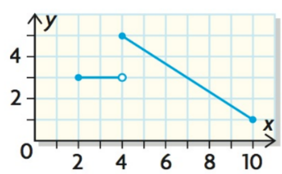
a)

x	*y*
−1	−3
0	1
1	5
2	9

b)

NOTES:

Function Test For Graphs: Use the **Vertical Line Test**. Imagine moving a ruler vertically across the graph. If the line/curve never passes through the ruler more than once anywhere, it passes the test and is a function



This feature of the graph is called a **hole**, represented by a hollow circle. This indicates that the function approaches the point (gets infinitely closer) but is *not defined at that precise value*

Communicating Functions Algebraically

Relations and functions may not always be expressed visually in a diagram or a chart. Sometimes there may be a rule which assigns an output value to y (the dependent variable) for each input value x (the independent variable). This rule may be expressed in different ways:

x	y
−1	−3
0	1
1	5
2	9

1. As an equation:

2. In function notation:

Function notation does not change the relationship in any way, it just allows for a standard way of communicating. Just as variables are interchangeable, the f in $f(x) = 4x + 1$ is just an identifier for the specific function; we could just as easily write $g(x) = 4x + 1$

<u>Ex. 2</u>: Consider the relationship between inches and feet.

a) How many inches are in a foot?

b) Create an *<u>equation</u>* to convert the number of inches into the number of feet:

Write the equation using variables that best represent inches and feet

c) Since the number of feet is a ***function of the number of inches***, write in function notation:

d) Evaluate f(36). What does this mean?

Notice that this notation can be applied to any variables, equation, or context. We could just as easy create an equation for the conversion between pounds (p) and kilograms (k):

$p(k) = 2.2k$ since there are ~2.2lbs in every kg

or pounds and ounces (z):

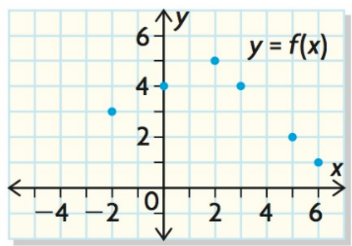
$p(z) = 0.0625z$

<u>Ex. 3:</u> Evaluate *f(-2)* for each of the following

a) f = { (4,0), (-1,9), (-2,1), (0,12) }

b)

<u>You Try It:</u> GoFormative Q3-Q4



Portfolio Assignment: Mathematical Stories

This portfolio replaces traditional homework assignments with a curated collection of mathematical stories. The goal is to show how your understanding develops over time: through struggle, error analysis, explanation, and extension. You will build this portfolio throughout the module (approximately two weeks) and then discuss it in a one-on-one 10-minute conference.

A story is a short, focused narrative showing how your understanding of a specific type of math problem or concept developed, supported by concrete evidence (questions, work, checks, AI transcripts, and reflection).

This portfolio assesses your reasoning, verification, communication, and conceptual understanding, not speed or memorization. Visual design is not graded; clarity of thinking and evidence is.

Use at least three (3) problems from this module's homework (the assigned textbook questions) to create three mathematical stories...

Story 1:

Purpose: Using at least one (1) textbook question, show how your understanding changed.

Required Components:

- **Initial Attempt (Struggle)**
  - A type of problem you initially found difficult
  - Include an image of the question as well as your original answer and work (consider one from class formative work, but a problem from the web or the textbook works too)
- **Revised Understanding (Breakthrough)**
  - A *new* question or sequence of questions that tests the **same underlying concept** (not just similar difficulty)
  - Include an image and your answer and work for each
- **Reflection**
  - What did I misunderstand at first?
  - What helped me see it differently?
  - What do I now understand about this concept that I didn't before?

Cheat Code: If you struggle early in this module, that's not a problem — that's actually Story 1 waiting to happen!

Start with a formative question from class that you honestly tried. After you've used my feedback, keep that feedback in mind while jumping into similar homework questions. Do this for each lesson and you'll eventually see at least one of the process click. When you do, notice what changes in how you think.

If you can tell the story of what didn't make sense at first and what clicked later, that's an authentic growth story — and that's exactly what I'm looking for.

Linear vs. Quadratic Functions

How can you tell if a relationship is linear or quadratic?

The language used depends on the representation:

<u>Algebraic (an equation):</u>

<u>Graphical:</u>

<u>Table of Values:</u>

<u>Ex. 4:</u> Identify each relation as linear or quadratic:

a) $f(x) = 4x - 1$

<u>You Try it</u>: Answer Formative Q5 & Q6 (Multiple Choice)

Consolidation

Answer Q7 in today's Formative (M1L1)

You have until 10am tomorrow to complete it, at which point I will provide you feedback and post the solution(s) to the announcements